

Iron and Total Iron-binding Capacity (TIBC)

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Percent of saturation; serum iron; total iron binding capacity; unsaturated iron binding capacity

Use

Differential diagnosis of anemia, especially with hypochromia and/or low MCV. The percent saturation sometimes is more helpful than is the iron result for iron deficiency anemia. Evaluate thalassemia and possible sideroblastic anemia; work-up hemochromatosis, in which iron is increased and iron saturation is high. Decrease in iron level after performance of Schilling supports the diagnosis of vitamin B₁₂ deficiency, vide infra. Evaluate iron poisoning (toxicity) and overload in renal dialysis patients, or patients with transfusion dependent anemias. Use of TIBC in iron toxicity may be less useful than previous believed.¹ TIBC or transferrin is a useful index of nutritional status.

Uncomplicated iron deficiency: Serum transferrin (and TIBC) high, serum iron low, saturation low. Usual causes of depleted iron stores include blood loss, inadequate dietary iron. RBCs in moderately severe iron deficiency are hypochromic and microcytic. Stainable marrow iron is absent. Serum ferritin decrease is the earliest indicator of iron deficiency if inflammation is absent.

Anemia of chronic disease: Serum transferrin (and TIBC) low to normal, serum iron low, saturation low or normal. Transferrin decreases with many inflammatory diseases. With chronic disease there is a block in movement to and utilization of iron by marrow. This leads to low serum iron and decreased erythropoiesis. Examples include acute and chronic infections, malignancy and renal failure.

Sideroblastic anemia: Serum transferrin (and TIBC) normal to low, serum iron normal to high, saturation high.

Hemolytic anemias: Serum transferrin (and TIBC) normal to low, serum iron high, saturation high.

Hemochromatosis: Serum transferrin (and TIBC) slightly low, serum iron high, saturation very high.

Protein depletion: Serum transferrin (and TIBC) may be low, serum iron normal or low (if patient also is iron deficient). This may occur as a result of malnutrition, liver disease, renal disease (eg, nephrosis) or other entities.

Liver disease: Serum transferrin variable; with acute viral hepatitis, high along with serum iron and ferritin. With chronic liver disease (eg, cirrhosis), transferrin may be low. Patients who have cirrhosis and portacaval shunting have saturated TIBC/transferrin as well as high ferritin.²

Chronic dialysis for renal failure: monitor iron levels in patients undergoing dialysis. To follow treatment of iron overload with deferoxamine or with regimen of recombinant human erythropoietin and phlebotomy.³

Limitations

Ferritin levels are also useful for iron deficiency. Low iron level may not indicate iron deficiency in acute infection with leukocytosis. Low iron levels may be misleading in chronic infection, inflammation and malignancy; high ferritin levels occur in many such states, however, the most sensitive test for iron deficiency is bone marrow examination.⁴ TIBC and transferrin are increased in patients on oral contraceptives, with normal saturation. Gross hemolysis may interfere with serum iron.

Contraindications

Parenteral iron before sample is drawn will cause misleading high iron results. Recent blood transfusion may have only a small positive effect on iron.

Custom Additional Information

Serum iron is increased in hemosiderosis, hemolytic anemias especially thalassemia, sideroachrestic anemias, hepatitis, acute hepatic necrosis, hemochromatosis, and with inappropriate iron therapy. Iron may reach high levels with iron poisoning. Some patients who receive multiple transfusions (eg, some hemolytic anemias, thalassemia, renal dialysis patients) will have increased serum iron levels.

Serum iron is decreased with insufficient dietary iron, chronic blood loss (including the hemolytic anemias paroxysmal nocturnal hemoglobinuria), inadequate absorption of iron and impaired release of iron stores as in inflammation, infection and chronic diseases. The combination of low iron, high TIBC and/or transferrin and low saturation indicates iron deficiency. Without all of these findings together, iron deficiency is unproven.² Low ferritin supports the diagnosis of iron deficiency.

Detection of iron deficiency may lead to detection of adenocarcinoma of gastrointestinal tract, a point which cannot be overemphasized. In recovery from pernicious anemia, especially just after B₁₂ dose, iron levels are low. In fact, the drop in serum iron 1 to several days after the Schilling test flushing dose of vitamin B₁₂ may be more useful in diagnosis than the radioactivity of the 24-hour urine collection. Serum iron is reported to drop with acute infarct of myocardium.

TIBC is increased in iron-deficiency, use of oral contraceptives, and in pregnancy.

TIBC is decreased in hypoproteinemia due to many causes, and is decreased in a number of inflammatory states.

Increased saturation occurs with HLA-related (classical) hemochromatosis before ferritin is greatly increased, and also with iron overload (eg, cirrhosis and portacaval shunt), in hemolytic anemias and with iron therapy. Saturation >70% in females, >80% in males is described as prerequisite for parenchymal loading; however, sample contamination and the vagaries of fluctuation in serum iron levels can make such criteria misleading on occasion.²

The serum ferritin is a more sensitive test than the serum iron or TIBC for iron deficiency and for iron overload.² When all these tests are used together, as is often necessary, they usually can distinguish between iron deficiency anemia and the anemia of chronic disease. The best and most reliable evaluation of total body iron stores is by bone marrow aspiration and biopsy. The best evaluation of iron deficiency in childhood (unless lead toxicity is suspected) is free erythrocyte porphyrins.

With recombinant erythropoietin therapy serum iron, transferrin saturation, and ferritin levels decline due to rapid utilization by stimulated erythropoiesis with resultant decrease in storage iron.^{3,5}

While iron is usually considered in relation to hematopoiesis and oxygen transport functions of red cells, it is also of prime import to the lymphomyeloid systems.⁶

Specimen Requirements

Specimen

Serum (preferred) **or** plasma (lithium heparin only). EDTA plasma causes decreased results.

Volume

2 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Storage Instructions

Maintain specimen at room temperature.

Patient Preparation

Have sample drawn before patient is given therapeutic iron or blood transfusion. Iron determinations on patients who have had blood transfusions should be delayed for at least four days.

Causes for Rejection

Hemolysis; specimen clotted; improper labeling; EDTA or oxalate plasma specimens

Footnotes

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2. Finch CA, Huebers H. Perspectives in iron metabolism. *N Engl J Med*. 1982 Jun 24; 306(25):1520-1528 (review). PubMed 7043270 (<http://www.ncbi.nlm.nih.gov/pubmed/7043270>)
3. McCarthy JT, Johnson WJ, Nixon DE, Jenson BM, Moyer TP. Transfusional iron overload in patients undergoing dialysis: Treatment with erythropoietin and phlebotomy. *J Lab Clin Med*. 1989 Aug; 114(2):193-199. PubMed 2754306 (<http://www.ncbi.nlm.nih.gov/pubmed/2754306>)
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5. Gordeuk VR, Brittenham GM, Hughes M, Keating LJ, Oppllt JJ. High-dose carbonyl iron for iron deficiency anemia: A randomized double-blind trial. *Am J Clin Nutr*. 1987 Dec; 46(6):1029-1034. PubMed 3318377 (<http://www.ncbi.nlm.nih.gov/pubmed/3318377>)
6. deSousa M, Brock JH. *Iron in Immunity, Cancer and Inflammation*. New York, NY: John Wiley & Sons;1989.